**Skillbit Technologies**

Internship Project Report

# Iris Flower Classification

Using Random Forest Classifier

Domain: Machine Learning

Date: June 04, 2026

## 1. Objective

The objective of this project is to build a machine learning classification model that can accurately predict the species of an Iris flower based on its physical measurements — sepal length, sepal width, petal length, and petal width. The model classifies flowers into three species: **Setosa**, **Versicolor**, and **Virginica**.

## 2. Dataset Details

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| **Dataset Name** | Fisher's Iris Dataset |
| **Source** | sklearn.datasets.load_iris() — Built-in |
| **Total Samples** | 150 samples (50 per class) |
| **Classes** | Setosa, Versicolor, Virginica |
| **Features** | 4 — Sepal Length, Sepal Width, Petal Length, Petal Width |
| **Class Balance** | Balanced — 50 samples per class |
| **Feature Units** | Centimeters (cm) |

## 3. Algorithm — Random Forest Classifier

Random Forest is an ensemble learning algorithm that constructs multiple decision trees during training and outputs the class selected by the majority of trees. It reduces overfitting compared to a single decision tree and provides high accuracy on classification tasks.

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| **n_estimators** | 100 trees |
| **max_depth** | 5 |
| **random_state** | 42 |
| **Criterion** | Gini Impurity (default) |

## 4. Methodology

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| **Step 1** | **Data Loading**: Loaded Iris dataset using sklearn.datasets.load_iris(). Converted to Pandas DataFrame for analysis. |
| **Step 2** | **EDA**: Plotted histograms for each feature grouped by species. Generated pairplot to visualize feature relationships. |

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| Step 3 | **Preprocessing**: Applied StandardScaler to normalize features. Performed stratified 80/20 train-test split. |
| Step 4 | **Model Training**: Trained Random Forest Classifier with 100 estimators on the training set. |
| Step 5 | **Evaluation**: Evaluated using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix. |
| Step 6 | **Prediction**: Demonstrated live prediction on a new flower sample with confidence scores. |

## 5. Results & Performance

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| **93.33%**<br>Overall Accuracy | **0.93**<br>Macro Precision | **0.93**<br>Macro Recall |
| **0.93**<br>Macro F1-Score | **120**<br>Training Samples | **30**<br>Test Samples |

**Per-Class Performance:**

| Class | Precision | Recall | F1-Score | Support |
|---|---|---|---|---|
| Setosa | 1.00 | 1.00 | 1.00 | 10 |
| Versicolor | 0.90 | 0.90 | 0.90 | 10 |
| Virginica | 0.90 | 0.90 | 0.90 | 10 |
| **Weighted Avg** | 0.93 | 0.93 | 0.93 | 30 |

## 6. Key Findings

- Petal Length and Petal Width are the most discriminative features for classification.
- Setosa species is perfectly separable from Versicolor and Virginica.
- Versicolor and Virginica show slight overlap, making them harder to distinguish.
- Random Forest with 100 estimators achieves 93.33% accuracy on unseen test data.
- StandardScaler normalization improved model stability and convergence.

## 7. Technology Stack

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| **Language** | Python 3.x |

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| **ML Library** | scikit-learn (RandomForestClassifier, StandardScaler, metrics) |
| **Data** | pandas, numpy |
| **Visualization** | matplotlib, seaborn |
| **IDE** | Visual Studio Code |

## 8. Conclusion

The Iris Flower Classification project successfully demonstrates the application of the Random Forest Classifier for multi-class classification. The model achieved an overall accuracy of 93.33% on the test dataset, with perfect classification of the Setosa species. The project covered the complete ML pipeline — from data loading and exploratory data analysis to model training, evaluation, and live prediction. This project provides a strong foundation for understanding supervised learning and ensemble methods in machine learning.